VDT AND IDT

CIRCUIT ANALYSIS METHOD

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GRADING SYSTEM

FINAL GRADE

30%

PRELIM

30%

MIDTERM

40%

ENDTERM

TERM GRADE

50%

WEEKLY

50%

EXAM

100%

LABORATORY

PENALTY

0

ABSENT

3

45-min LATE

7 max

15-min LATE

TOPIC OUTLINE

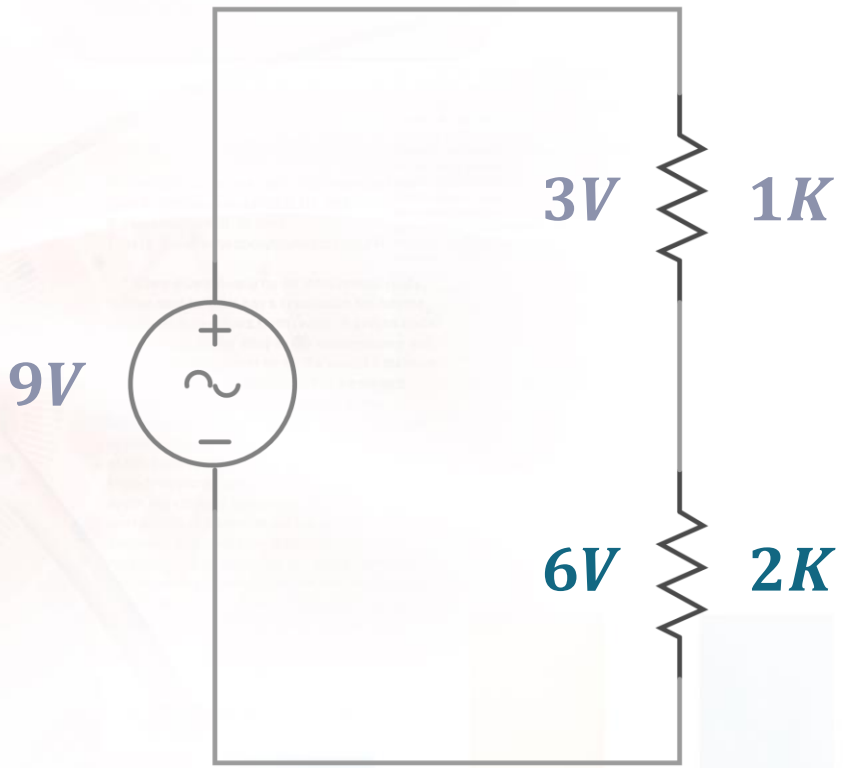
Voltage Divider Theorem

Current Divider Theorem

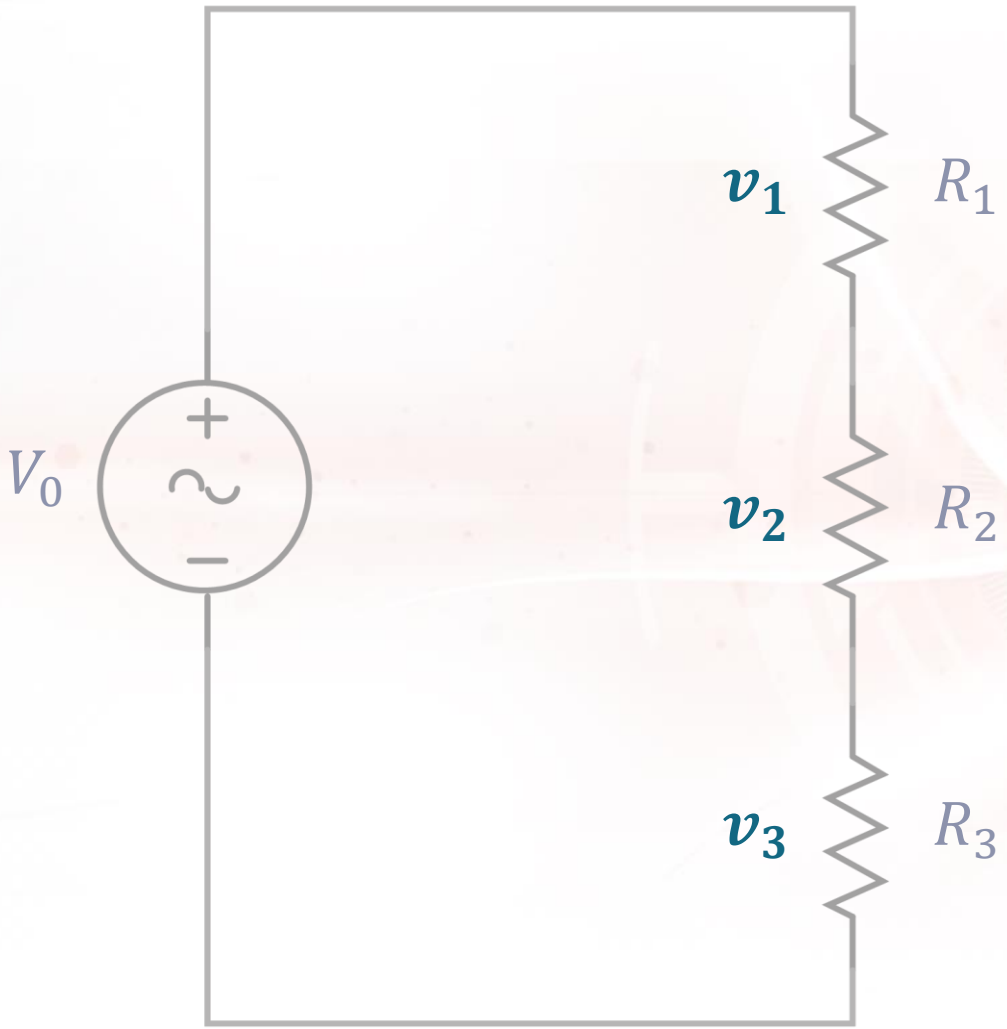
VOLTAGE DIVIDER THEOREM

VOLTAGE DIVIDER THEOREM

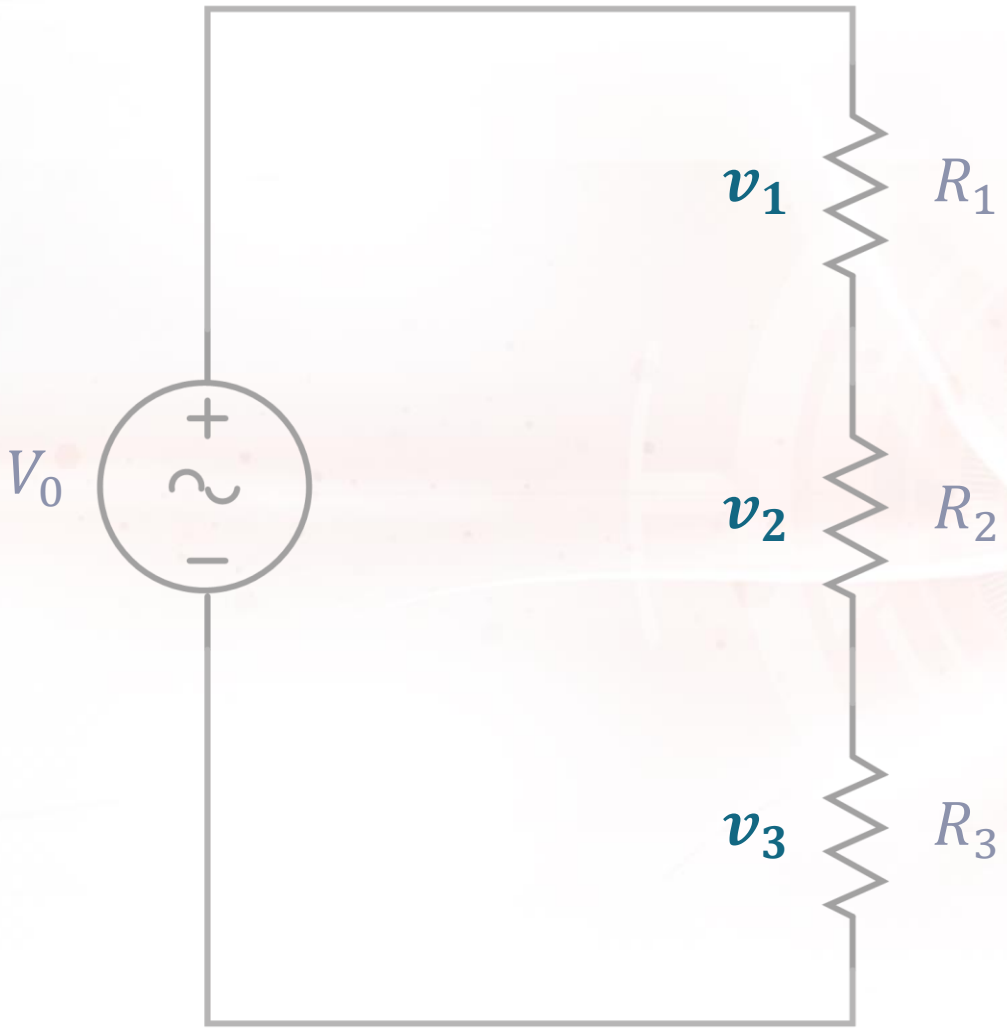
In a **series circuit**, with multiple resistors, the **voltage** across each resistor is **proportional** to its resistance relative to the total resistance on the circuit



VDT FORMULA DERIVATION



VDT FORMULA



$$v_1 = V_O \frac{R_1}{R_O}$$

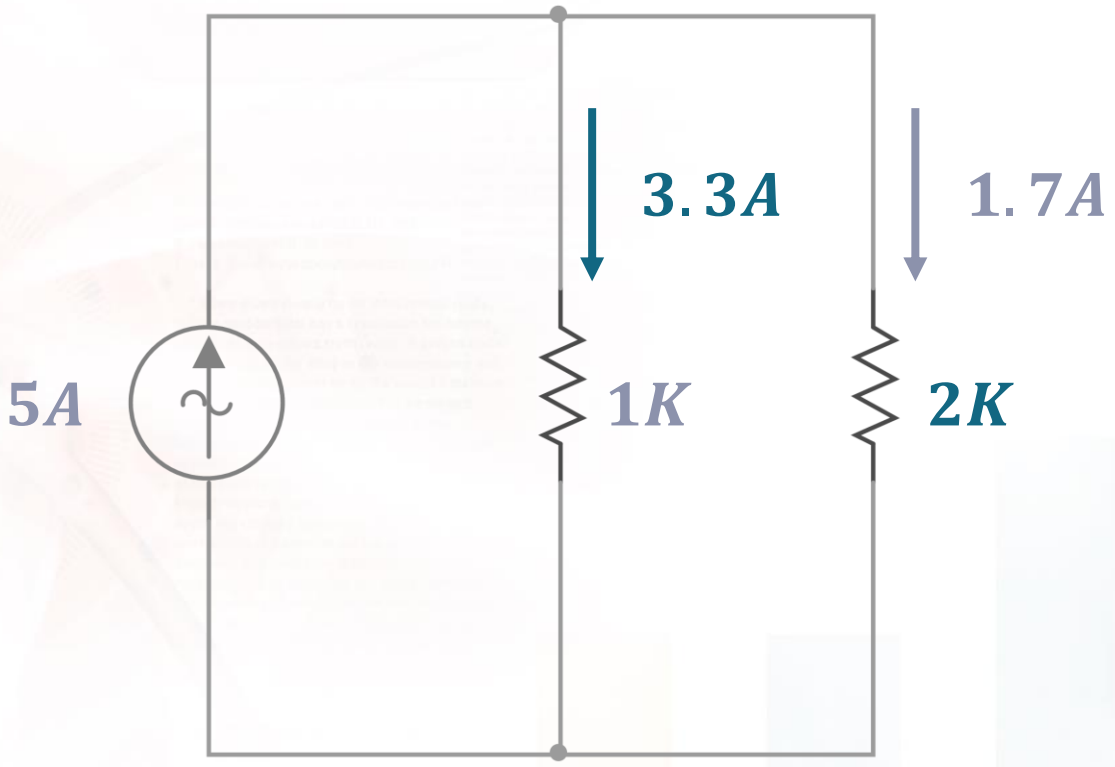
$$v_2 = V_O \frac{R_2}{R_O}$$

$$v_3 = V_O \frac{R_3}{R_O}$$

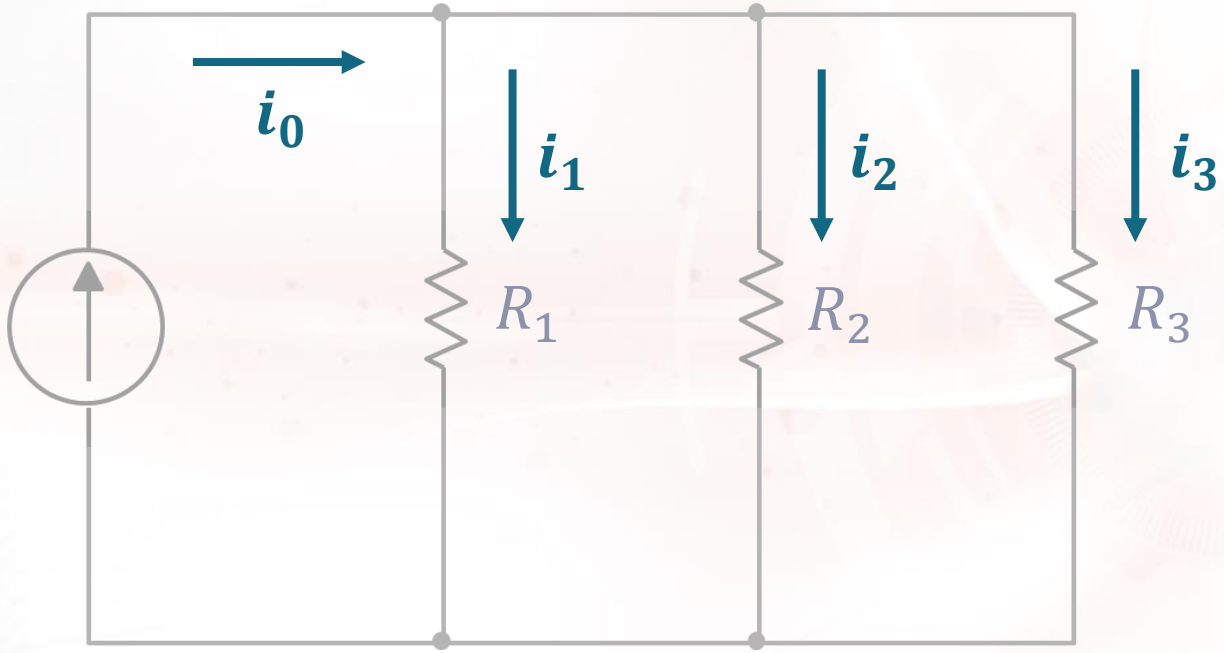
CURRENT DIVIDER THEOREM

CURRENT DIVIDER THEOREM

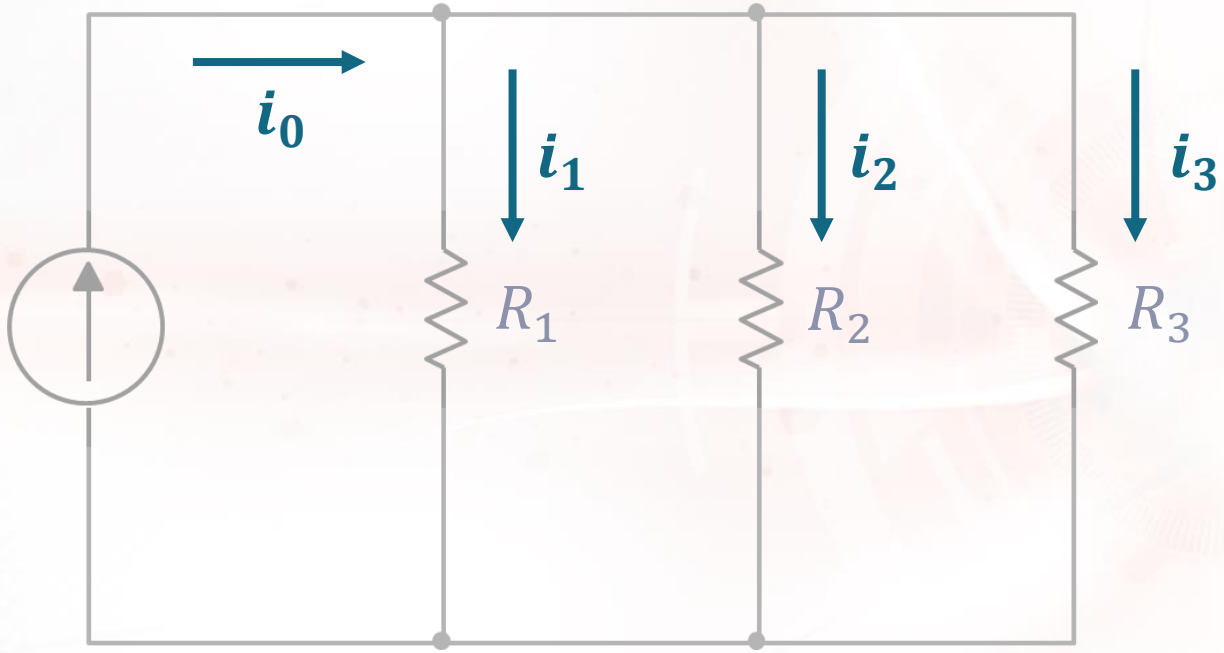
In a **parallel circuit**, with multiple resistors, the **current** through each resistor is **inversely proportional** to its resistance relative to the total resistance on the circuit



IDT FORMULA DERIVATION



IDT FORMULA

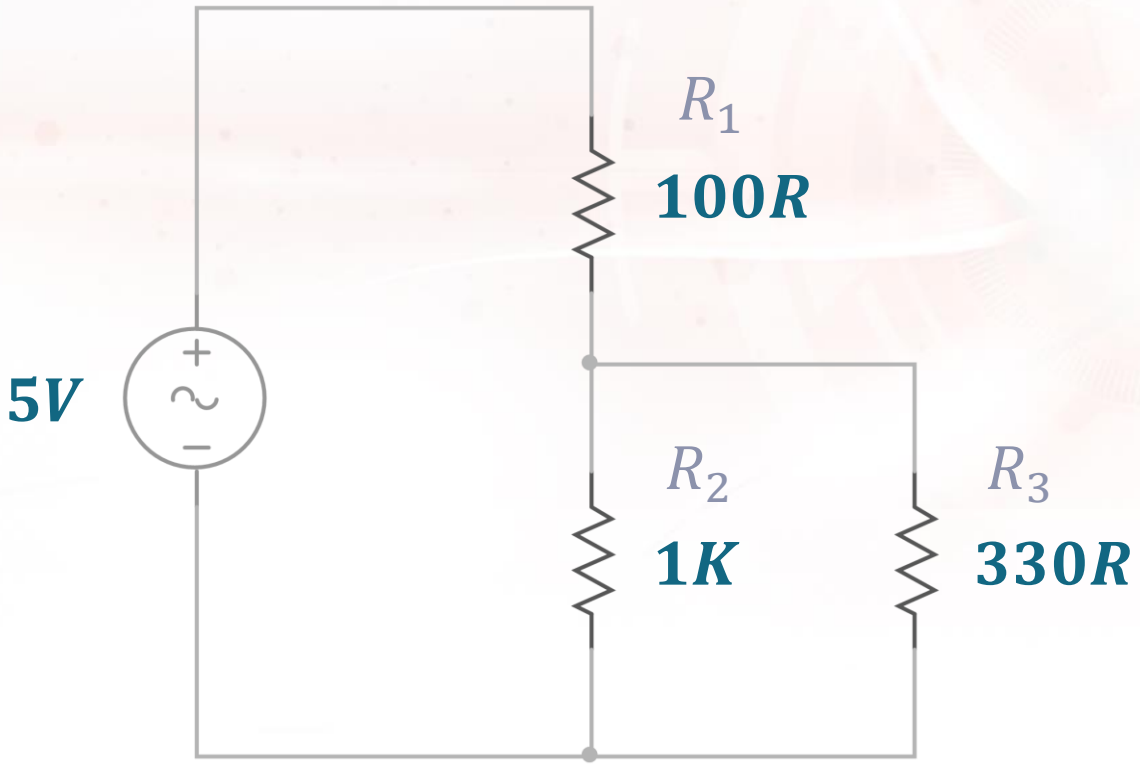


$$i_1 = I_O \frac{R_O}{R_1}$$

$$i_2 = I_O \frac{R_O}{R_2}$$

$$i_3 = I_O \frac{R_O}{R_3}$$

Determine the value of **current**, **voltage**, and **power** of every element in the given circuit using **VDT/IDT**.



LABORATORY